

QUADAS-3: A Revised Tool for the Quality Assessment of Diagnostic Test Accuracy Studies

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Assessing risk of bias and applicability is critical in diagnostic test accuracy reviews to ensure valid, relevant findings that can inform decision making and policy. This article introduces QUADAS-3, an updated version of the widely used QUADAS-2 tool that was developed in response to the latest evidence, feedback from users of QUADAS-2, and comprehensive piloting.

The QUADAS-3 tool comprises 6 phases: 1) define the systematic review synthesis question, 2) define the ideal test accuracy trial for each synthesis question, 3) draw a flow diagram, 4) identify estimates of accuracy for assessment, 5) assess risk of bias and applicability, and 6) make an overall judgment. Risk of bias is assessed across 4 domains: Participants, Index Test, Target Condition, and Analysis. The first 3 domains are also assessed in terms of concerns regarding applicability to the synthesis question.

Signaling questions are included to help judge risk of bias. The QUADAS-3 tool incorporates several changes compared with the previous version (QUADAS-2) which will ideally improve its validity, usability, and usefulness. Key changes from QUADAS-2 include introduction of the concept of an ideal test accuracy trial, assessment at the estimate level rather than the study level, replacement of the "Flow and Timing" domain with a new "Analysis" domain, introduction of sections on the rationale for judgments, and formal introduction of overall judgment of risk of bias and concerns regarding applicability. Some changes to domains and signaling questions have also been made.

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Assessing risk of bias and applicability is essential in systematic reviews of test accuracy to ensure the validity and relevance of the findings. Flaws in the design, conduct, and analysis of test accuracy studies can lead to bias (that is, overestimation or underestimation of test accuracy) (1), ultimately affecting the care of patients and the public. Applicability is also important because differences in study populations, index tests, and reference standards relative to the review question can limit the generalizability of findings to different settings. Systematically evaluating both risk of bias and applicability enhances confidence in the rigor and relevance of test accuracy reviews for supporting informed decision making, policy development, and guideline recommendations.

The QUADAS-2 tool (2), published in 2011, was designed to evaluate the risk of bias and applicability of test accuracy studies included in systematic reviews using signaling questions across 4 domains: patient selection, index test, reference standard, and flow and timing. The tool has been widely cited and is an update of the original QUADAS tool that was published in 2003, moving from a simple checklist to a domain-based

tool (3). QUADAS-2 is recommended by the Cochrane Collaboration and used by major Health Technology Assessment organizations (4, 5).

Although user feedback on QUADAS-2 has generally been positive, some signaling questions were deemed problematic. In addition, both the development process for QUADAS-C (6) for the assessment of comparative test accuracy studies and the "Assessing Risk of Bias and Applicability" chapter of the Cochrane Handbook for Systematic Reviews of Diagnostic Test Accuracy identified areas for improvement (5). These developments motivated an update of the QUADAS-2 tool. There was also potential for the tool to benefit from changes implemented in recently developed risk-of-bias tools for other study types (7, 8). In this article, we describe QUADAS-3, an update of the QUADAS-2 tool that is based on the latest evidence, feedback from users of QUADAS-2, and comprehensive piloting.

DEVELOPMENT OF QUADAS-3

The process of developing QUADAS-3 began in 2019 but was interrupted by the COVID-19 pandemic. The project resumed in 2022. We established a core working group of 7 members, with an additional 6 members forming the QUADAS-3 Steering Group (coauthors of this article; see the **Appendix**, available at *Annals.org*). We followed a 3-step process to develop QUADAS-3 (Table 1).

See also:

Web-Only

Related Explanation and Elaboration article

Supplemental material

Step 1: Evidence to Inform QUADAS-3

We followed a series of stages to identify evidence on potential changes to be considered for the QUADAS-3 tool. These are summarized in Table 1; a more detailed overview of the methods and results for each stage is provided in Supplement 1 (available at Annals.org). The core group collated and summarized the suggested changes from each of the steps. Suggestions were classified as changes supported by the core group, changes requiring further discussion or piloting, or changes deemed unnecessary by the core group. This informed the development of the first draft of QUADAS-3 (version 0.2) for discussion at the face-to-face meeting (step 2).

Step 2: Face-to-Face Meeting and Scope

The face-to-face meeting was held on 2 September 2023 in London, England, before the Cochrane Colloquium. Twelve of the 13 members of the steering group attended. We discussed the results from step 1 and how to incorporate these into QUADAS-3, helping to inform the scope. We discussed how to consider applicability as part of the updated tool and decided to retain applicability as part of QUADAS-3 but agreed that a clearer structure and additional guidance were needed. We worked in small groups to consider potential changes to domains based on the draft produced by the core group (version 0.2), with each group producing an initial updated version of the domains allocated to them. We also explored whether it would be feasible to develop algorithms to derive domain-level risk-of-bias judgments, but after discussion among groups and exploration of potential options for doing so, it was agreed that this would not be feasible without compromising the usability of the tool.

Step 3: Piloting and Refinement

After the face-to-face meeting, the core group incorporated the suggestions into a revised version of QUADAS-3 (version 0.4). This was shared with the rest of the steering group for further comments and refinement. We held regular meetings of the core group from September 2022 to April 2025, discussing any issues identified with the tool and how to address them. As part of this process, we also explored whether we should integrate QUADAS-C (6) into QUADAS-3 to create a single tool.

We undertook a series of piloting exercises and refined the tool after each exercise (9). Members of the steering group piloted the tool (version 0.6) on a set of 5 papers during July 2024, with each paper being assessed by 3 to 5 members of the group. The papers were selected to cover a range of index tests, target conditions, and methodological issues. This process identified areas for improvement which were incorporated into the next version of the tool. It also provided us with a set of QUADAS-3 assessments with model answers to help users see how the tool should be applied in practice. These exemplars were refined

and updated to the final version of QUADAS-3 once piloting was complete. They can be accessed from the QUADAS website (www.quadas.info).

A draft of QUADAS-3 (version 0.8) was piloted at a workshop during the Global Evidence Summit (September 2024) with 16 participants, most of whom were familiar with QUADAS-2. Feedback was generally positive, and no major changes were required. A revised version (version 0.9) was shared with the steering group and refined to version 0.11. This was used for formal think-aloud piloting. Six reviewers with varying experience applied the tool to a paper on a dipstick test for diagnosing typhoid (10), verbalizing their thoughts as they worked through the assessment. Feedback informed version 0.12, which was further refined by the core group. Seven reviewers piloted

Table 1. Overview of Methods Used to Develop QUADAS-3

Project Step	Overview
Step 1: Identification of evidence to inform the development of QUADAS-3	Regular meetings of core group to compile and assess evidence, including: A. Modifications to QUADAS-2 for the Cochrane Handbook B. Developments from more recent risk-of-bias tools, including QUAPAS (10), QUADAS-C (6), PROBAST (9), RoB 2 (8), and ROBINS-I (7) C. Review of 3 methodological studies that evaluated QUADAS-2 D. Review of 66 systematic reviews published in 2019 that used QUADAS-2 E. Web-based survey of 53 review authors who have applied QUADAS-2 F. Review of guidance on use of QUADAS or QUADAS-2 from 6 HTA organizations that recommended these tools
Step 2: Face-to-face meeting of steering group	Regular meetings of core group to develop first draft of QUADAS-3 Meeting of the steering group to discuss: Findings of step 1 Scope of QUADAS-3 How to address applicability Potential changes to domains Feasibility of algorithms for arriving at domain-level ratings Very early draft of tool (version 0.2)
Step 3: Piloting and refinement	Regular meetings of core group to develop successive iterations of the tool Revised version of tool created and shared with steering group for feedback (version 0.4) Consideration of incorporation of QUADAS-C Piloting of tool on 5 papers by the steering group (version 0.6) Piloting of tool at Global Evidence Summit 2024 (version 0.8) Revised tool piloted in a series of think-aloud interviews (version 0.11) Revised tool piloted in “real” systematic reviews (version 0.13) Revised tool shared with steering group for comments (version 0.14) Final version agreed on by steering group (version 1.0)

HTA = Health Technology Assessment; PROBAST = Prediction model study Risk Of Bias Assessment Tool; RoB 2 = Cochrane Risk of Bias 2 tool; ROBINS-I = Risk Of Bias In Non-randomised Studies - of Interventions.

Table 2. Overview of the Phases Included in the QUADAS-3 Tool

Phase	When to Complete
Phase 1: State the systematic review synthesis questions	Once per review
Phase 2: Define the ideal test accuracy trial for each synthesis question	Once per review
Phase 3: Draw a flow diagram	Once per study
Phase 4: Identify the accuracy estimates to assess for risk of bias and applicability	Once per study
Phase 5: Assess risk of bias and concerns regarding applicability	For each selected estimate
Phase 6: Overall judgment	For each selected estimate

the tool in 5 real-life reviews. These were either completed reviews in which the reviewers had previously used QUADAS-2 or ongoing reviews where the reviewers agreed to pilot QUADAS-3. Feedback via a web-based survey was largely positive, with only minor, nonstructural suggestions. The tool was updated after these changes were made and shared with the steering group (version 0.14) for final comments. Piloting also identified areas needing clarification, which informed the accompanying explanation and elaboration guidance on how to use QUADAS-3 (11).

During the piloting, we explored the option of moving from “signaling questions” to “signaling statements.” There was no strong preference for either version of the tool, and the core group decided to keep the version with questions.

THE QUADAS-3 TOOL

The full QUADAS-3 tool is available in **Supplement 2** (available at Annals.org) and on the QUADAS website (www.quadas.info). The tool is designed to assess the risk of bias and applicability of primary test accuracy studies for diagnosis, screening, or staging of disease. The tool is to be used in 6 phases. Phases 1 and 2 are conducted at the review level, phases 3 and 4 at the study level, and phases 5 and 6 at the accuracy estimate level. **Table 2** provides an overview of the phases included in the QUADAS-3 tool, the point in the assessment at which each section should be completed, and which part of the tool should be completed for each phase.

We present a brief overview of each phase in the following sections. Full details on how to apply the tool are provided in the accompanying explanation and elaboration document (11).

Phase 1: State the Systematic Review Synthesis Question

A single review may address more than 1 synthesis question. The first phase of the QUADAS-3 assessment is to specify with sufficient detail (population, index test, and target condition) the synthesis question or questions that will be answered in the systematic review; this should also be prespecified in the systematic review protocol.

Phase 2: Define the Ideal Test Accuracy Trial for Each Synthesis Question

For each synthesis question defined in phase 1, the ideal test accuracy trial that could be conducted to answer this question should be defined. QUADAS-3 provides a generalized framework and design template to prompt and help users to define the ideal test accuracy trial for each synthesis question (**Table 3**). The ideal test accuracy trial design should minimize the risk of bias and maximize the applicability of the trial to each synthesis question. It is important to note that the ideal test accuracy trial is a theoretical concept. This is the ideal test accuracy trial that could be conducted to answer the synthesis question, and it is unlikely that studies included in a review will match this design due to ethical and resource constraints. Deviations from the ideal test accuracy trial design are frequent and sometimes unavoidable. However, explicit consideration of deviations from the ideal design can inform judgments of risk of bias and applicability concerns that may distort study results from the true accuracy of the test in its proposed role in a testing pathway.

Phase 3: Draw a Flow Diagram

Review authors should draw a diagram to show the flow of participants through the study and to identify if the study includes estimates of test accuracy from different participant groups or other variations in study design. The flow diagram will provide information about the method of recruiting participants (for example, using a consecutive series of participants with specific symptoms who are suspected of having the target condition, or separate case and control participants), the order of test execution, and the number of participants undergoing the index test and the reference standard to facilitate judgments about risk of bias. A hand-drawn diagram is sufficient, as this step does not need to be reported as part of the QUADAS-3 assessment.

Phase 4: Identify the Accuracy Estimates to Assess Risk of Bias and Applicability

For each synthesis question, it is often possible to calculate multiple estimates of sensitivity and specificity (that is, different 2×2 tables) from the data presented in a study. Multiple estimates within a specific study may occur due to differences between study subgroups or index tests; index tests with multiple thresholds; or variations in the definitions of the target condition, reference standards, and units of analysis (for example, samples, lesions, or participants). The potential risk of bias and applicability issues may differ for different estimates, so review authors need to clearly identify the estimate for which risk of bias and applicability are being assessed.

Review authors are asked to first decide which estimates from a study will be assessed with QUADAS-3. Not all estimates reported in a primary study will be relevant; review authors should extract and assess only

those estimates that are relevant to one of the prespecified review synthesis questions (defined in phase 1) and that will contribute to the synthesis (whether narrative or statistical). Phase 4 includes completion of a structured table to identify features related to the estimate being assessed (for example, population, index test, or reference standard). Phase 5 assessments should be done for each set of estimates from a study included in the review. To avoid duplication when multiple estimates are being included in the review, we suggest that as part of this phase, review authors consider which of the 4 domains from phase 5 may have different risk of bias or applicability concerns for each estimate. Review authors can then complete the full assessment of all phase 5 domains for the first estimate and only need to assess domains that could have different risk of bias or applicability concerns for each subsequent estimate.

Phase 5: Assess Risk of Bias and Concerns Regarding Applicability

Phase 5 consists of 4 domains: Participants, Index Test, Target Condition, and Analysis. All 4 domains are assessed in terms of risk of bias, and the first 3 domains are also assessed in terms of concerns regarding applicability.

Signaling questions are included to help judge risk of bias; these questions flag aspects of study design, conduct, and analysis related to the potential for bias and aim to help reviewers judge risk of bias. Table 4 provides an overview of the phase 5 domains and signaling questions.

Signaling questions are answered as “yes” (Y), “probably yes” (PY), “probably no” (PN), “no” (N), or “no information” (NI). Risk of bias is judged as “low,” “high,” or “insufficient information.” If all signaling questions for a domain are answered “Y” or “PY,” risk of bias can be judged as “low.” If any signaling question is answered “N” or “PN,” this flags the potential for bias and review authors need to use their judgment and review-specific guidance to judge risk of bias. The “insufficient information” category should be used only when insufficient data are reported to permit a judgment. Concern regarding applicability to the synthesis question is judged as “low,” “high,” or “insufficient information.”

Phase 6: Overall Judgment

The final phase is to judge the overall risk of bias and concerns regarding applicability of each accuracy estimate across the 4 domains, which can be “low,” “high,” or “insufficient information.” If all domains are judged as “low” for risk of bias, the overall judgment can be considered to be “low”; if 1 or more domains are judged as having “high” risk of bias, the overall judgment should be “high”; and if 1 or more domains are judged as having “insufficient information” and no domains are judged as having “high” risk of bias, the overall judgment should be “insufficient information.” The same process applies to concerns regarding applicability. Review authors are also asked to provide

a rationale for the overall judgment of both risk of bias and concerns regarding applicability. This should be based on the rationale for each of the individual domain-level judgments.

DISCUSSION

This third iteration of the QUADAS tool was developed based on feedback from users and systematic reviews that have used QUADAS-2, developments in more recent risk-of-bias tools, methodological studies that have evaluated QUADAS-2, and modifications made to QUADAS-2 for Cochrane. We carried out

Table 3. Framework to Define the Ideal Test Accuracy Trial

Component	General Features
Overall	Prespecified protocol Sufficient total sample size to expect adequate numbers of participants with and without the target condition to generate sufficiently precise estimates of test accuracy (e.g., sensitivity and specificity)
Objective	Assess the accuracy of the index test in its proposed role and position in the clinical pathway
Participants	Prospective enrollment of a single group of consecutive participants or a random group of eligible participants suspected of having the target condition Participants are a representative sample of the intended-use population, as defined in the synthesis question Details of symptoms and prior test results of included patients reported
Index test	Index test should be the appropriate technical version, conducted using recommended instructions and in the intended role in the clinical pathway Performed after recruitment of trial participants Interpretation of the index test result should not be affected by external information that would not usually be available to the person interpreting the test in its proposed role in practice (e.g., information on the reference standard, disease state, or additional test results) Test conducted by operators who would usually conduct and interpret the test in practice, in a setting that is representative of the review question Definition of thresholds to classify test results as positive or negative prespecified
Definition of the target condition	Reference standard and reference standard threshold should correctly classify all participants as having or not having the target condition Protocol for undergoing the reference standard is the same for all study participants, and the reference standard is performed in all participants Reference standard performed after recruitment of trial participants Reference standard interpretation is independent of the index test; those interpreting the reference standard should not be aware of the index test results, and the index test should not be part of the reference standard when a composite reference standard is used The time between the index test and the reference standard is sufficiently short to ensure confidence that the disease status has not changed
Analysis	All participants included in the analysis Unit of analysis appropriate for the synthesis question Appropriate methods used to calculate estimates of accuracy (e.g., sensitivity and specificity)

Table 4. Overview of Phase 5 Domains and Signaling Questions

Domain	Participants	Index Test	Target Condition	Analysis
A. Risk of bias				
Information used to answer signaling questions	Describe how participants were enrolled (e.g., single-gate [a group of participants in whom the diagnosis was not yet known; this could include multiple groups from different locations, such as different hospitals] or multiple-gate [case-control design; participants with known diagnosis]) Describe whether enrollment was prospective or retrospective Describe how participants were selected for enrollment (consecutive or random sample) Describe any restrictions to enrollment	Describe the index test Describe whether the index test was conducted and interpreted according to recommended instructions Describe what information was available to those interpreting the index test If a threshold was used, describe how it was selected	Define the target condition Describe how the target condition was assessed (what reference standard was used and how it was conducted and interpreted) What was the interval between the index test and the reference standard?	Describe any participants who were enrolled in the study but excluded from the 2 × 2 table (refer to flow diagram) and reasons for their exclusion How were any missing 2 × 2 data, including missing reference standard data, accounted for? What do the numbers in the 2 × 2 table correspond to (i.e., what is the unit of analysis)? Describe any nonstandard statistical analyses used to generate the estimates of sensitivity and specificity being assessed (by either the study authors or you)
Signaling questions (yes, probably yes, no, probably no, or no information)	1.1: Was a single-gate design used? 1.2: Were participants prospectively enrolled? 1.3: Was a consecutive or random sample of participants included? 1.4: Is the study group a representative sample of the intended-use population?	2.1: Was the index test conducted and interpreted according to the recommended instructions? 2.2: Were the index test results interpreted without knowledge of the reference standard results? 2.3: Were the index test results interpreted with the same information as would be available when the test is used in practice? 2.4: If an index test threshold was used, was it either standard or prespecified?	3.1: Does the reference standard adequately identify those with and without the target condition? 3.2: Was the target condition assessed in all participants? 3.3: Was the target condition assessed in the same way in all participants? 3.4: Did the reference standard avoid incorporating the index test? 3.5: Was the reference standard conducted and interpreted according to the recommended instructions? 3.6: Were the reference standard results interpreted without knowledge of the index test results? 3.7: If a reference standard threshold was used, was it standard or prespecified? 3.8: Was there an appropriate time interval between index test and reference standard?	4.1: Were all participants included in the analysis? 4.2: Were missing data handled appropriately? 4.3: Does the unit of analysis match the ideal test accuracy trial? 4.4: Were the estimates of sensitivity and specificity calculated appropriately?
Risk of bias (high, low, insufficient information)	Risk that the selection of participants has introduced bias	Risk that the conduct or interpretation of the index test has introduced bias	Risk that the assessment of the target condition has introduced bias	Risk that the analysis has introduced bias
B. Concerns regarding applicability to the systematic review synthesis question				
Concerns regarding applicability (high, low, insufficient information)	Concern that the included participants do not match those in the ideal test accuracy trial	Concern that the index test, its conduct, or its interpretation does not match the ideal test accuracy trial	Concern that the target condition as defined by the reference standard does not match the ideal test accuracy trial	-

extensive piloting and discussions among the core group and the steering group to further develop and refine the revised tool.

The QUADAS-3 tool incorporates several changes compared with the previous version (QUADAS-2) which we hope will improve its validity, usability, and usefulness. These are summarized in **Table 5** and are discussed in the following sections. All changes are supported by an expanded explanation and elaboration document, which includes worked examples to facilitate review authors' use of the tool (11).

Structural Changes

A major change is that risk of bias and applicability concerns are now assessed at the accuracy estimate level rather than at the study level. Although this change may increase the time needed to complete a QUADAS-3 assessment, it is crucial because different estimates from the same study may have different potential for bias or concerns regarding applicability.

A related change is the possibility of defining multiple synthesis questions. This reflects the fact that, within a single review, one may be interested in more than 1 question. We have termed questions within a single review "synthesis questions" to reflect the fact that each separate question within a review should be addressed by a separate synthesis of results (study characteristics, quality assessment [risk of bias and concerns regarding applicability], and findings).

To aid with the assessment of concerns regarding applicability and to focus the risk-of-bias assessments, we have introduced the concept of the "ideal test accuracy trial." We ask review authors to define the ideal test accuracy trial that could be conducted in the intended-use setting to answer each prespecified synthesis question. We include a design template to help review authors define ideal test accuracy trials. Review authors are asked to refer to this ideal test accuracy trial when assessing concerns regarding applicability. The greater detail involved in specifying the ideal test accuracy trial provides a more helpful framework for the assessment of applicability than when the review question alone was used for assessing applicability concerns in QUADAS-2. Use of a broadly defined review question frequently caused challenges for the applicability assessment such that included studies with different applicability to different synthesis questions would all be judged as having the same level of concern.

A final structural change is the introduction of a formal section for judging overall risk of bias and applicability across domains. QUADAS-2 provided some guidance on this process but did not formally include a judgment of overall risk of bias and applicability. Our review of reviews and author survey highlighted that this was not done consistently.

Changes to Domains and Signaling Questions

We have made some more minor changes to the domain structure. The first 2 domains remain largely

unchanged, with only a change in wording from "patient selection" to "participants" for the first domain. The third domain has been renamed from "Reference Standard" to "Target Condition" to better reflect the intent of this domain: to determine whether participants are appropriately classified by the reference standard as having or not having the target condition. This explicit focus has resulted in our moving signaling questions concerned with verification bias and timing between the index test and the reference standard from the QUADAS-2 "Flow and Timing" domain to the new "Target Condition" domain. The "Target Condition" domain has been further expanded to capture additional sources of bias relating to the reference standard threshold and inclusion of the index test in the reference standard's definition. The "Flow and Timing" domain has been replaced by the "Analysis" domain to reflect its focus on missing data, the unit of analysis, and whether accuracy estimates are calculated appropriately.

At the domain level, we have included some new signaling questions in domain 1 on prospective enrollment of participants (question 1.1), in domain 2 on index test methods (question 2.1) and information available to those interpreting the test (question 2.3), and in domain 3 on conduct of the reference standard (question 3.2), the reference standard threshold (question 3.3), and incorporation bias (question 3.8). Domain 4 includes

Table 5. Overview of Changes From QUADAS-2 to QUADAS-3

Type of Change	Changes
Structural changes	Assessment of risk of bias and applicability concerns at the estimate level rather than the study level (phase 4) Definition of (multiple) synthesis questions (phase 1) Introduction of the ideal test accuracy trial to aid with the assessment of applicability and to focus the risk-of-bias assessments (phase 2) Formal section for overall judgment (phase 6)
Changes to domains and signaling questions	"Patient selection" renamed "Participants" "Reference standard" renamed "Target condition" (resulted in moving signaling questions from flow and timing related to target condition) "Flow and timing" domain replaced by "Analysis" Addition of new signaling questions and rewording of more challenging questions Answers to signaling questions: addition of "probably yes" (PY) and "probably no" (PN) options Domain-level judgments: replace "unclear" with "insufficient information" Introduction of rationale for judgment Phrasing of domain-level judgments as statements rather than questions Focus of applicability judgments on synthesis question and ideal test accuracy trial
Changes considered but not introduced	Use of algorithms Integration of QUADAS-C Move from signaling questions to statements

new signaling questions to allow for complex methods that are being developed to account for potential biases in the estimation of test accuracy, such as imperfect reference standards and whether missing data have been handled appropriately. We have also included new sections on “rationale for judgment” for all domain-level judgments on risk of bias and concerns regarding applicability. Review authors are asked to briefly summarize the key concerns that have led to a particular judgment. If they have answered 1 or more signaling questions as “no” but do not believe this will have resulted in bias, this should be explained here.

We have made changes to the wording of signaling questions highlighted as problematic in QUADAS-2 (step 1E [Table 1]) to improve clarity. We have expanded the signaling question response options to include “probably yes” and “probably no” and replaced “unclear” with “no information.” This brings the signaling questions in line with other tools, including the Cochrane Risk of Bias 2 tool, ROBINS-I (Risk Of Bias In Non-randomised Studies – of Interventions), and PROBAST (Prediction model study Risk Of Bias Assessment Tool), and aims to guide review authors away from a “no information” answer, which is less useful than a “yes” or “probably yes” answer or a “no” or “probably no” answer. We made minor changes to the description of the domain-level judgments, replacing the “unclear” option with “insufficient information.” This is in line with changes to signaling question responses to encourage judgment about the presence or absence of bias unless there is really not enough information to make a judgment.

We phrased the domain-level judgments as statements rather than questions so that the judgments of high, low, or insufficient information follow more clearly. For example, the participants domain now includes the statement, “Risk that the selection of participants has introduced bias,” which previously read, “Could the selection of patients have introduced bias?” Similarly, for the applicability judgments, we now have the statement, “Concern that the included participants do not match those in the ideal test accuracy trial,” which previously read, “Is there concern that the included patients do not match the review question?” We also relate the assessment of concerns regarding applicability much more clearly to the synthesis question and the ideal test accuracy trial defined for each synthesis question. This is to ensure that review authors know they should be assessing applicability to the synthesis question defined specifically for their reviews, not applicability in a more general sense.

Changes Considered but Not Implemented

There were several changes that we considered but decided not to implement. The main one was the use of algorithms to perform domain-level assessments. We received feedback that this would be helpful for review authors, and this was explored in detail during the face-to-face meeting of the steering group at the

start of the project (step 2). We concluded that it was not possible to develop domain-level algorithms without considerably increasing the complexity of the tool in terms of the numbers of signaling questions and links between questions. The steering group believed that this increased complexity would discourage use of the tool, so we agreed not to proceed with algorithms. We also explored integrating QUADAS-3 and QUADAS-C into a single tool. It was agreed that because most reviews and primary studies are still noncomparative, this would lead to a tool that would be unnecessarily complex for the majority of users. Instead, we will develop a separate specific update of QUADAS-C. The explanation and elaboration document (11) includes guidance on how to use the current version of QUADAS-C alongside QUADAS-3. Finally, we considered moving from signaling questions to signaling statements. This was explored as part of the piloting, but because there was no clear preference for either version of the tool, the core group decided to retain the more familiar “signaling questions.”

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